

TRANSPORTATION & LOGISTICS ANALYTICS REPORT

Exploratory Data Analysis, SQL Database Design, and Power BI

PROJECT SCOPE

6,773 cleaned shipment records | 32 analyzed features | March 2019 - September 2020

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1. Executive Summary

This end-to-end analytics project investigates why shipments were delayed, where operational bottlenecks were concentrated, and how a connected data model can support faster intervention. The workflow combines data cleaning, exploratory analysis, SQL Server database design, Power BI reporting, and business recommendations.

Metric	Result
Cleaned shipment records	6,773
Delayed shipments	4,263 (62.94%)
On-time shipments	2,510 (37.06%)
Analysis period	March 2019 - September 2020
Crisis months	10 months at 96-100% delay rate

PRIMARY CONCLUSION

Nearly two in every three shipments missed the planned ETA. The pattern is systemic and is associated with route distance, tracking quality, vehicle type, partner performance, and incomplete master data.

Business Question

Why are shipments delayed, where are the main operational bottlenecks, and what controls should management prioritize?

Project Deliverables

- A cleaned and feature-engineered shipment dataset.
- A reproducible Python EDA notebook.
- A 13-table SQL Server database proof of concept with 10 analytical queries.
- A three-page Power BI dashboard for delay, partner, fleet, and GPS analysis.
- A prioritized operational recommendation plan.

2. Dataset and Methodology

Dataset Overview

Attribute	Description
Raw records	6,880
Cleaned records	6,773
Features analyzed	32
Primary date field	Planned ETA
Delay definition	Actual ETA > Planned ETA

Cleaning and Feature Engineering

- Standardized column names, data types, and date fields.
- Removed invalid or unusable records and reviewed missing-value patterns.
- Standardized vehicle, supplier, customer, and GPS-provider labels where possible.
- Created delivery status, year, month, year-month, day-of-week, distance segment, vehicle class, body type, and route-location features.
- Validated shipment totals and reconciled delayed and on-time classifications.

Analytical Workflow

Stage	Purpose
Python EDA	Profile data quality and identify performance drivers
SQL Server	Model connected logistics operations and reusable queries
Power BI	Turn findings into interactive management views
Recommendations	Translate evidence into operational actions

3. Key Findings: Delay and Data Quality

Systemic Delay Exposure

A total of 4,263 shipments were delayed, producing an overall delay rate of 62.94%. Ten monthly periods within the July 2019 to May 2020 window reached delay rates between 96% and 100%; April 2020 has no shipment records.

Delay driver	Evidence
Very long routes	81.08% delay rate
Short routes	73.08% delay rate
Manual GPS tracking	99% delay rate
VAMOSYS tracking	12% delay rate

Data Quality Blind Spots

- 737 shipments (10.9%) were assigned to an unknown supplier.
- 751 shipments had an unknown vehicle type.
- More than 30 GPS-provider labels indicate limited standardization.
- Incomplete master data weakens accountability and can distort partner comparisons.

INTERPRETATION

The delay problem cannot be solved by a single blanket policy. It requires controls that distinguish route risk, tracking quality, vehicle assignment, and partner performance.

4. Key Findings: Operational Drivers

Area	Finding	Business implication
Vehicle type	32 FT single-axle 7MT: 81.53% delayed	Review dispatch, maintenance, and assignment rules
High-volume vehicle	40 FT 3XL trailer 35MT: 80.81% delayed across 2,559 trips	High volume amplifies the operational exposure
Supplier	K. Ramachandran: 95.38% delay rate	Requires formal root-cause and SLA review
Customer	Ford India: 73.35% delay rate	Service performance should be monitored by account
Geography	Eastern destinations indicate potential empty-return exposure	Backhaul opportunities should be investigated

What the Findings Mean

- Delay exposure is concentrated rather than evenly distributed.
- Tracking-provider choice is associated with major performance differences.
- The highest-volume trailer category also carries substantial delay risk.
- Unknown supplier and vehicle records create avoidable management blind spots.
- The evidence supports targeted interventions instead of one universal operating rule.

5. Business Recommendations

Priority	Action	Expected management outcome
1	Control high-risk routes	Route-specific service rules, capacity plans, and exception alerts
2	Standardize GPS data	Reduce manual tracking and monitor provider-level performance
3	Intervene by vehicle type	Audit dispatch, maintenance, loading, and route assignment
4	Enforce partner accountability	Root-cause reviews and clear service-level ownership
5	Strengthen data validation	Mandatory supplier, vehicle, driver, and tracking identifiers
6	Use recurring management reviews	Monitor, investigate, act, and confirm KPI movement

OPERATING PRINCIPLE

Use the Power BI dashboard as a recurring exception-management system, not as a one-time report.

Recommended Control Loop

- Monitor: Power BI flags changes and exceptions.
- Investigate: SQL queries trace the route, partner, vehicle, or tracking driver.
- Act: Operations assigns ownership and corrective action.
- Review: KPI movement confirms whether the intervention worked.

6. SQL Database Solution

The proposed SQL Server layer turns disconnected logistics records into a traceable operational model. It is a designed solution and proof of concept: the historical shipment dataset supports the EDA and Power BI findings, while the demo transactional tables illustrate how a future live system could capture operational events.

Layer	Tables
Reference data	Countries, Shipment Status, Failure Reasons, Material
Business entities	Hubs, Customers, Employees, Couriers
Core transaction	Shipments
Connected operations	Tracking Events, Delivery Attempts, Payments, Proof of Delivery

Database Components

- 13 relational tables connected through primary and foreign keys.
- Bulk-load logic for the supplied CSV source tables.
- Reusable shipment-master and tracking-timeline views.
- 10 analytical queries covering shipment history, tracking status, hub throughput, courier success, failures, late and stuck shipments, proof of delivery, and revenue.

VALUE

The model establishes a single source of truth for operational monitoring and provides a structured foundation for Power BI and future predictive analytics.

7. Conclusion and Portfolio Links

The analysis shows that shipment delay was a systemic operational issue, but the risk was not evenly distributed. Route distance, GPS-provider quality, vehicle type, supplier performance, and incomplete master data define where management attention should be concentrated.

The project demonstrates an end-to-end analytics workflow: raw data was cleaned and explored in Python, structured through a relational SQL Server design, translated into interactive Power BI views, and converted into practical business recommendations.

Project Assets

Asset	Location
GitHub repository	github.com/Mohamedtarek00212/Transportation-Logistics
Live Power BI dashboard	Available from the repository README
LinkedIn	linkedin.com/in/mohamedtarek-aboelkhir

FINAL TAKEAWAY

A connected data foundation turns delayed-shipment analysis from a historical explanation into a repeatable management system with measurable ownership.

Technology Stack

Python | Pandas | NumPy | Matplotlib | Seaborn | SQL Server | Power BI | Excel | PowerPoint